

Exam 1 Prep

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Loading data

- `install.packages()` comes from Base R and it installs packages to be used in R. Ex) `install.packages("Dplyr")`.
- `library()` comes from Base R and loads packages. Must use in document before trying to use data or functions that are in certain packages. Ex) `library(tidyverse)`.

Getting Information on Data -all of the following come from the Base R package unless otherwise specified

- `max()` returns the largest value.
- `mean()` returns the average value.
- `min()` returns the smallest value.
- `names()` gets or sets the names of elements. Ex) `names(df)`
- `ncol()` returns the number of variables in a data set.
- `nrow()` returns the number of rows in a data set.
- `sum()` adds values together.
- `summarize()` comes from the dplyr package and returns a data frame of summary statistics.
- `c` combines values to make one vector. Ex) `subjects <- c("math", "english", "science")`

ggplots -all of the following come from the ggplot2 package

- `ggplot()` creates a plot
- `aes()` maps variables to visual properties of a plot. Ex) `ggplot(df, aes(x=, y= )) + geom_point()`
- `geom_bar()` gives the ggplot bars. This is better for categorical variables.
- `geom_density()` creates a density plot, a smoothed version of a histogram.
- `geom_histogram()` creates a histogram with x variables being numeric.
- `geom_line()` adds a line to the ggplot that connects the points.
- `geom_point()` adds the data points to the plot.
- `geom_smooth()` adds a trend line to the plot, with a confidence interval.
- `labs()` adds labels to the plot. Ex) `labs(title = "", x = "", y = "")`.

Reshape and Transform Data -all of the following come from the dplyr package unless otherwise specified

- `mutate` creates new columns or modifies existing ones. Ex) `mutate(df, new_column = some_calculation)`
- `pivot_longer()` comes from the tidyr package and it transforms data to the “long” or “hot dog” format.
- `pivot_wider()` comes from the tidyr package and it transforms data to the “wide” or “hamburger” format.
- `arrange()` sorts the rows of a variable(s) (column) in ascending order. Ex) `arrange(variable)`
- `count()` counts the number of rows of a column/dataset as groups of some variable.
- `desc()` is often used inside of the `arrange()` function where it will sort the rows of variable(s) but in descending order. Ex) `arrange(desc(variable))`
- `distinct()` removes duplicate rows of a data set.
- `filter()` filters or keeps the rows that meet a certain condition. Ex) `filter(df, income>$10k)`
- `group_by()` organizes data by groups. Ex) `group_by(df, condition)`
- `head()` comes from the Base R package and it shows the first few rows of a data set.

- `relocate()` moves the columns to a new position, the specified ones go to the beginning by default. Ex) `relocate(df, variable1, variable2)`
- `rename()` changes the specified column names without touching the data in the column. Ex) `rename(new_name = old_name)`
- `select()` reorders or sorts the columns, keeping the ones we tell it to and getting rid of the rest of them.
- `dim()` comes from the Base R package and it tells us the number of rows and columns in a data set.
- `starts_with()` selects columns whose name starts with _____. This is often used inside of `select`, `rename`, and `relocate`.